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**NOVO PROTOCOLO PARA AVALIAR "IN VITRO" A MICROINFILTRAÇÃO
MICROBIANA ATRAVÉS DOS CIMENTOS ENDODÔNTICOS**

Nova Friburgo

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Dissertação apresentada ao Programa de Pós-Graduação em Odontologia do Instituto de Saúde de Nova Friburgo da Universidade Federal Fluminense, como parte dos requisitos para obtenção do título de Mestre em Odontologia, na área de concentração: Clínica Odontológica.

Orientadora: Professora Natalia Iorio Lopes Pontes Póvoa

Coorientador: Professor Leonardo dos Santos Antunes

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“Você nunca sabe a força que tem. Até que a sua única alternativa é ser forte.”

(Johnny Depp)

RESUMO

O tratamento endodôntico visa reduzir ao máximo a carga microbiana presente no sistema de canais radiculares por meio do preparo químico-mecânico, utilizando instrumentos manuais e/ou mecanizados sob ação de substâncias irrigadoras. Para evitar o fracasso do tratamento endodôntico, é importante selar adequadamente os canais após o preparo químico-mecânico, a fim de evitar microinfiltração microbiana. O selamento envolve o uso de guta-percha e cimento endodôntico, este último aderindo à dentina e preenchendo os espaços entre a guta-percha e o sistema de canais radiculares, entretanto falhas na integridade e limitações dos materiais obturadores e restauradores podem predispor a microinfiltração. Com o objetivo de se entender melhor a microinfiltração, estudos laboratoriais apresentam diversos tipos de análise em materiais obturadores. Entre os modelos, destaca-se o método de duas câmaras para avaliar a infiltração microbiana, contudo não garante que esta microinfiltração ocorre através do espaço do canal radicular e não por outras vias. O objetivo deste estudo foi desenvolver um protocolo laboratorial para avaliar a microinfiltração microbiana através dos cimentos endodônticos, de forma a garantir a visualização da via de infiltração do inóculo. Para tanto, foi desenvolvido um sistema contendo raízes de dentes bovinos, tubos de vidro com pérolas de vidro e meio de cultura ágar bile esculina. Cada raiz, após preparo, desinfecção e esterilização em autoclave, foi inserida em um tubo estéril de vidro contendo pérolas de vidro, previamente pesadas, este conjunto foi parcialmente preenchido com meio ágar bile esculina e sua esterilidade verificada pela incubação por 24 h. Após este período, o sistema foi inoculado com *Enterococcus faecalis* ATCC 29212, incubado a 36°C e observado por 6 semanas. O ensaio laboratorial proposto permitiu detectar se a infiltração por *E. faecalis* ocorreu via canal radicular ou externamente a raiz devido a alteração da coloração por oxidação do meio de cultura; na região apical através do seu longo eixo do canal radicular ou superficialmente ao meio de cultura quando, por falha do operador, alguma quantidade do inóculo seja depositado externamente à raiz. Conclui-se que este novo protocolo apresenta as características necessárias para avaliar microinfiltração microbiana através dos cimentos endodônticos, visto sua capacidade de demonstrar a via da microinfiltração, garantindo diferenciar microinfiltração de contaminação.

Palavras-chave: bactéria; microbiologia; endodontia; infiltração dentária; protocolo.

ABSTRACT

Endodontic treatment aims to reduce as much as possible the microbial load present in the root canal system through chemical-mechanical preparation, using manual and/or mechanized instruments under the action of irrigating substances. To avoid the failure of endodontic treatment, it is important to adequately seal the canals after chemical-mechanical preparation, to prevent microbial microleakage. The sealing involves the use of gutta-percha and endodontic cement, the latter adhering to dentin and filling the spaces between the gutta-percha and the dentin walls; however, failures in integrity and limitations of obturating and restorative materials may predispose to microleakage. With the objective of better understanding microleakage, laboratory studies present several types of analysis in obturating materials. Among the models, the two-chamber method stands out for evaluating microbial leakage; however, it does not guarantee that this microleakage occurs through the root canal space and not by other pathways. The objective of this study was to develop a laboratory protocol to evaluate microbial microleakage through endodontic sealers, in order to ensure visualization of the inoculum leakage pathway. For this purpose, a system was developed containing bovine tooth roots, glass tubes with glass beads, and bile esculin agar culture medium. Each root, after preparation, disinfection, and sterilization in autoclave, was inserted into a sterile glass tube containing previously weighed glass beads; this set was partially filled with bile esculin agar medium and its sterility verified by incubation for 24 h. After this period, the system was inoculated with *Enterococcus faecalis* ATCC 29212, incubated at 36°C, and observed for 6 weeks. The proposed laboratory assay allowed detection of whether infiltration by *E. faecalis* occurred via the root canal or externally to the root due to color change by oxidation of the culture medium; in the apical region through the long axis of the root canal or superficially to the culture medium when, due to operator failure, some amount of the inoculum was deposited externally to the root. It is concluded that this new protocol presents the necessary characteristics to evaluate microbial microleakage through endodontic cements, given its ability to demonstrate the pathway of microleakage, ensuring differentiation between microleakage and contamination.

Keywords: bacteria; microbiology; endodontics; dental leakage; protocol.

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LISTA DE ABREVIATURAS E SIGLAS

APC	<i>Article Publication Charge</i> (Taxa de Publicação de Artigo)
ATCC	<i>American Type Culture Collection</i> (Coleção Americana de Culturas de Tipo)
BESH	<i>Basic experimental studies in humans</i> (Estudos experimentais básicos em humanos)
BHI	<i>Brain Heart Infusion</i> (Infusão Cerebral Cardíaca)
CAPES	Coordenação de Aperfeiçoamento de Pessoal de Nível Superior
CCDC	<i>Cambridge Crystallographic Data Centre</i> (Centro de Dados Cristalográficos de Cambridge)
CFU	<i>Colony Forming Units</i> (Unidades Formadoras de Colônia)
COPE	<i>Committee on Publication Ethics</i> (Comitê de Ética em Publicações)
CRedit	<i>Contribution Roles Taxonomy</i> (Taxonomia de Papéis de Contribuição)
DDBJ	<i>DNA Data Bank of Japan</i> (Banco de Dados de DNA do Japão)
EDTA	<i>Ethylenediaminetetraacetic Acid</i> (Ácido Etilenodiaminotetracético)
FAPERJ	Fundação Carlos Chagas Filho de Amparo à Pesquisa do Estado do Rio de Janeiro
IACUC	<i>Institutional Animal Care and Use Committee</i> (Comitê Institucional de Cuidados e Uso de Animais)
MTA	<i>Mineral Trioxide Aggregate</i> (Agregado de Trióxido Mineral)
NaOCl	<i>Sodium Hypochlorite</i> (Hipoclorito de Sódio)
PIR	<i>Protein Information Resource</i> (Recurso de Informações sobre Proteínas)
PPE	<i>Personal Protective Equipment</i> (Equipamento de Proteção Individual)
PQM	Preparo Químico Mecânico
RRIDs	<i>Research Resource Identifiers</i> (Identificadores de Recursos de Pesquisa)
SAPs	<i>Statistical Analysis Plans</i> (Planos de Análise Estatística)
SCR	Sistema de Canais Radiculares
SisGen	Sistema Nacional de Gestão do Patrimônio Genético e do Conhecimento Tradicional Associado
UFF	Universidade Federal Fluminense

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1 INTRODUÇÃO

A perda da integridade das estruturas dentárias favorece a penetração de microrganismos no sistema de canais radiculares (SCR). Esse evento ocorre quando estruturas dentárias, como o esmalte ou cemento, que compõem a parte externa do dente, e a dentina, que representa a maior parte da estrutura dentária, sofrem ruptura por fatores biológicos ou traumáticos (Farci; Soni, 2023). O estudo clássico de Kakehashi, Stanley e Fitzgerald (1965), demonstrou que a presença de microrganismos é determinante para o desenvolvimento de lesões periapicais, o que foi corroborado por estudos mais recentes (Aw, 2016; Farci; Soni, 2023; Nair, 2004).

Dentre os microrganismos frequentemente associados à infecção endodôntica destacam-se *Enterococcus faecalis* e *Candida albicans* (Ajmi *et al.*, 2024), que possuem a capacidade de formar biofilmes, conferindo-lhes resistência aos processos de limpeza do SCR, elevando o risco de falha no tratamento endodôntico (Siqueira Jr; Rôças, 2008). Particularmente, *E. faecalis* destaca-se por sua resistência aos antimicrobianos convencionais e pela habilidade de sobreviver em condições desfavoráveis dentro do SCR (Sharma *et al.*, 2025). A complexa estrutura dos canais radiculares, incluindo istmos e canais laterais, se tornam nichos para a sobrevivência microbiana, dificultando a limpeza química e mecânica (Ajmi *et al.*, 2024; Martins *et al.*, 2019; Prada *et al.*, 2019; Sharma *et al.*, 2025; Siqueira Jr; Rôças, 2008).

Nesse contexto, o tratamento endodôntico visa reduzir ao máximo a carga microbiana presente no SCR por meio do preparo químico-mecânico (PQM), utilizando instrumentos manuais e/ou mecanizados sob ação de substâncias irrigadoras, como hipoclorito de sódio (Haapasalo *et al.*, 2010), e obturação com materiais biocompatíveis (Gatelli; Bortolini, 2014; Macedo *et al.*, 2021; Schilder, 2006; Vier *et al.*, 2011).

A guta-percha, que atua como material obturador, apresenta ação limitada, e não é capaz, isoladamente, de garantir um selamento duradouro (Schilder, 2006). Dessa forma, a utilização de cimentos endodônticos, pode contribuir para a melhora desse selamento (Komabayashi *et al.*, 2020; Kowalski *et al.*, 2024). Sendo assim, a guta-percha associada aos cimentos endodônticos buscam preencher de forma tridimensional o SCR, auxiliando na prevenção de futuras infecções (Hori *et al.*, 2021; Siqueira Jr.; Rôças; 2008).

Evitar a microinfiltração microbiana, através da obturação, é essencial para aumentar a eficácia, a previsibilidade e o sucesso dos procedimentos endodônticos (Ajmi *et al.*, 2024; Jafari *et al.*, 2017). Contudo, falhas na integridade e limitações dos materiais obturadores e restauradores podem resultar em microinfiltração (Ajmi *et al.*, 2024; Jafari; Jafari, 2017; Wuerschling *et al.*, 2023).

Para que sejam considerados ideais, os cimentos endodônticos devem reunir propriedades que garantam excelente selamento e compatibilidade biológica com os tecidos periapicais (Rekha *et al.*, 2023). Esses cimentos podem apresentar características como: biocompatibilidade, selamento eficaz, ação antimicrobiana, estabilidade em meio úmido, radiopacidade, boa adesão ao canal radicular, tempo de trabalho prolongado e ausência de contração durante a presa (Branstetter; Von Fraunhofer, 1982), e podem apresentar em sua composição: óxido de zinco-eugenol, salicilato, óxido de zinco-ácido graxo, ionômero de vidro, silicone, resina epóxi, silicato tricálcico (agregado de trióxido mineral MTA/biocerâmica) e resina de metacrilato (Komabayashi *et al.*, 2020).

Ainda que sejam encontrados cimentos endodônticos com as propriedades descritas anteriormente, tem-se observado o insucesso dos tratamentos endodônticos devido à microinfiltração microbiana, resultando em uma infecção endodôntica secundária (D'costa; Bangerá; Kutty, 2017; Khayat; Lee; Torabinejad, 1993; Roghanizad; Jones, 1996; Swanson; Madison, 1987).

É importante realizar pesquisas laboratoriais que verifiquem a microinfiltração através dos cimentos endodônticos já utilizados, bem como também de novos (Akhavan *et al.*, 2011; Jafari; Jafari, 2017; Jafari *et al.*, 2017). Assim sendo, estudos apresentam diferentes metodologias para verificar a microinfiltração, como, por infiltração de glicose, com corantes como o azul de metileno, por pressão de fluidos no corpo de prova (calculado a partir do movimento da bolha) e também microinfiltração bacteriana; com o intuito de qualificar o selamento dos condutos radiculares (Akhavan *et al.*, 2011; Jafari; Jafari, 2017; Jafari *et al.*, 2017). Dentre os modelos para verificar microinfiltração microbiana, destaca-se o método que utiliza um sistema de duas câmaras. A câmara superior é composta por um microtubo adaptado, contendo raízes dentárias previamente preparadas e seladas com esmalte de unha e cianoacrilato. Após esterilização com óxido de etileno, essa câmara é acoplada à inferior (um frasco de vidro contendo meio de cultura líquido estéril). A extremidade da raiz permanece submersa no meio, sem tocar o fundo do frasco. A turvação

do meio de cultura indica ocorrência de microinfiltração bacteriana (Mansour *et al.*, 2024; Shahi *et al.*, 2017; Yanpiset *et al.*, 2018).

No entanto, revisões de literatura apontam limitações nos estudos de microinfiltração microbiana com duas câmaras (Jafari; Jafari, 2017; Rechenberg; De-Deus; Zehnder, 2011), como o uso do cianoacrilato como material selante, onde a microinfiltração pode ocorrer na interface dente-tubo da câmara superior e não pelo espaço do canal radicular (Jafari *et al.*, 2017).

Embora muitos dos estudos que avaliaram microinfiltração tenham utilizado dentes humanos (Jafari *et al.*, 2017; Jafari; Jafari, 2017; Kim *et al.*, 2015; Komabayashi *et al.*, 2020; Moradi *et al.*, 2009; Rechenberg; De-Deus; Zehnder, 2011; Rekha *et al.*, 2023; Wuerschling *et al.*, 2023), a substituição por dentes bovinos em estudos de endodontia tem se mostrado eficaz (Azevedo *et al.*, 2025; Brackett *et al.*, 1998; Carvalho *et al.*, 2018; Schmalz *et al.*, 2001; Silva *et al.*, 2019; Yassen; Platt; Hara, 2011). Mesmo considerando que a dentina bovina apresenta maior permeabilidade, essa diferença é pequena, uma vez que o número e diâmetro de túbulos são semelhantes às da dentina humana, porém com maior densidade (Schilke *et al.*, 2000), sendo os dentes bovinos considerados uma alternativa viável para testes de permeabilidade e resistência de união em obturação com gutta-percha e cimento (Carvalho *et al.*, 2018; Silva *et al.*, 2019). Adicionalmente, a utilização desses dentes em ensaios laboratoriais apresenta como vantagem a maior facilidade de obtenção em comparação aos dentes humanos (Schmalz *et al.*, 2001), cuja disponibilidade é limitada (Tanaka *et al.*, 2008).

Uma vez que o selamento eficaz do SCR está associado ao sucesso do tratamento endodôntico e os métodos de avaliação da microinfiltração microbiana são questionados por não garantirem que a via de acesso do microrganismo ao meio de cultura ocorreu através da obturação, torna-se importante o desenvolvimento de um novo modelo experimental que permita avaliar com precisão a infiltração microbiana em raízes obturadas com diferentes materiais obturadores.

2 OBJETIVO

2.1 OBJETIVO GERAL

Desenvolver um novo modelo laboratorial para avaliação da microinfiltração microbiana pelos cimentos endodônticos, utilizando dentes bovinos, assegurando a identificação precisa da via de infiltração.

3 ARTIGO A SER SUBMETIDO AO PERIÓDICO *CURRENT PROTOCOLS*

Os resultados da pesquisa original realizada serão apresentados sob a forma de artigo científico, que será submetido ao periódico *Current Protocols*. As normas para submissão estão apresentadas no Apêndice A, no Apêndice B encontram-se o convite para submeter o trabalho a revista. O apêndice C apresenta o cadastro do acesso ao Patrimônio Genético no Sistema Nacional de Gestão do Patrimônio Genético e do Conhecimento Tradicional Associado (SisGen). A seguir, apresenta-se o artigo a ser submetido:

ARTICLE TITLE:

A Novel Laboratorial Protocol to Evaluate Bacterial Microleakage of Endodontic Sealers

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ABSTRACT:

To avoid failure of endodontic treatment, it is important to properly seal the canals after chemomechanical preparation, in order to avoid microleakage. Typically, sealing involves the use of gutta-percha and sealing cement, the latter of which adheres to the dentin and fills the spaces between the gutta-percha and the dentin walls. Laboratory studies have been performed to detect microleakage of endodontic sealers based on various criteria, including microbiological ones, however none guarantee that the leakage occurred through the root canal space and not through other pathways. Our protocols propose a new laboratory assay to evaluate bacterial microleakage in endodontic sealers, ensuring visualization of the correct inoculum infiltration pathway. Furthermore, we present the main critical parameters and provide suggestions on how to manage them. These protocols allow the detection of bacterial microleakage in bovine teeth filled with endodontic sealer and inoculated with *Enterococcus faecalis*.

Basic Protocol 1: Bovine teeth selection and cleaning

Basic Protocol 2: Bovine teeth inclusion

Basic Protocol 3: Tooth standardization

Basic Protocol 4: Set preparation

Basic Protocol 5: Inoculum preparation

Basic Protocol 6: Canal inoculation

Basic Protocol 7: Set incubation

KEYWORDS:

Bacteria, Microbiology, Endodontics, Dental Leakage, Protocol

INTRODUCTION:

The tooth has two main parts, crown and root, being externally covered by enamel and cement, respectively. Internally, it is composed of dentin and dental pulp (Farci & Soni, 2023). Endodontic infection occurs mainly due to the exposure of the dental pulp to the oral cavity (Nair, 2004), and the presence of microorganisms in the root canal system triggers an inflammatory response resulting in apical periodontitis (Aw, 2016; Kakerashi *et al.*, 1965; Nair, 2004).

The success of endodontic treatment in teeth with apical periodontitis is associated with reducing the bacterial load to levels compatible with healing (Siqueria & Roças, 2008). This treatment relies on a chemomechanical approach, which combines instrumentation and irrigation procedures for achieving effective cleansing of the root canal system (Byström *et al.*, 1987; Garcez *et al.*, 2007). To avoid endodontic treatment failure it is also important to properly seal the canals after chemomechanical preparation, in order to prevent microleakage (Deshpande & Naik, 2015; Jafari & Jafari, 2017; Komabayashi *et al.*, 2020; Muliya *et al.*, 2014).

Typically, sealing involves the use of gutta-percha, a semi-solid material, and sealing cement. Gutta-percha acts as a filling material, while endodontic cement adheres to the dentin and fills spaces between gutta-percha and the dentinal walls (Muliyar *et al.*, 2014).

Endodontic sealers are classified by composition as: zinc oxide-eugenol, salicylate, zinc oxide-fatty acid, glass ionomer, silicone, epoxy resin, tricalcium silicate (MTA Mineral Trioxide Aggregate/bioceramic), and methacrylate resin (Komabayashi *et al.*, 2020). Currently, different laboratory studies have been performed for detecting microleakage of endodontic sealers, based on: dye penetration, dye extraction, fluid filtration, glucose penetration, bacterial leakage, endotoxin infiltration, caffeine penetration, caffeine microleakage, protein penetration, copper ion diffusion, radioisotope penetration, and electrochemical leakage tests (Jafari & Jafari, 2017; Komabayashi *et al.*, 2020). Review studies have presented limitations and disadvantages associated with these methodologies (Jafari & Jafari 2017; Rechenberg *et al.*, 2010; Verissimo & Do Vale, 2006), especially regarding microbial leakage in two-chamber models. The main limitation they report is that this model does not guarantee that the leakage actually occurred through the root canal space pathways and not through other pathways (Rechenberg *et al.*, 2010).

Our study brings a novel laboratory assay to evaluate the bacterial microleakage of endodontic sealers (Fig. 1), ensuring visualization of the correct inoculum leakage pathway. This will allow identification of whether or not the leakage is occurring through the root canal space. We describe the selection and cleaning of bovine teeth in basic protocol 1. Basic protocol 2 presents the inclusion criteria and storage of the teeth. Once teeth are included, the decoronation process and standardization of shape and size of the root canals are described in basic protocol 3, whereas basic protocol 4 provides the preparation of a set containing the tooth to be inoculated. The inoculum preparation and the set inoculation with *Enterococcus faecalis* are described in basic protocol 5 and 6, respectively. Basic protocol 7 demonstrates how to incubate and maintain the sets throughout all 6 weeks of experiment.

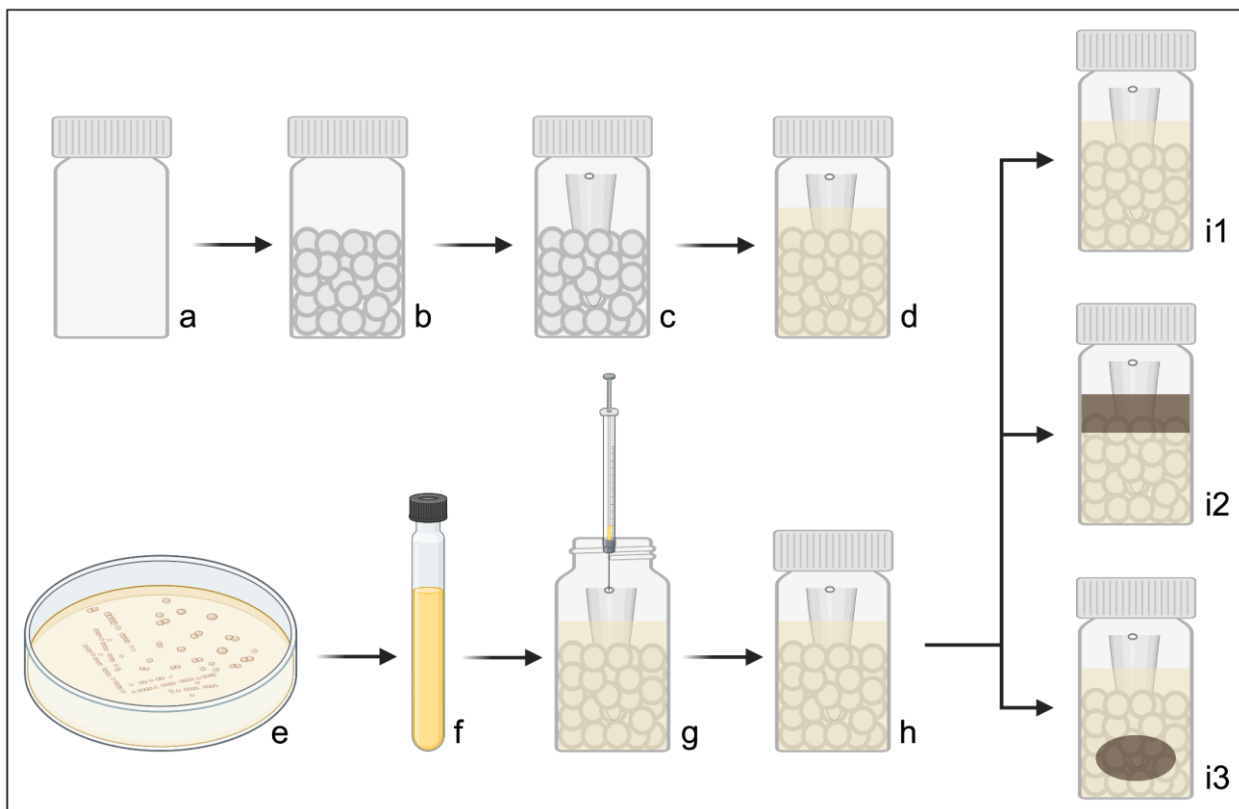


Figure 1: Novel laboratory protocol to evaluate the bacterial microleakage of endodontic sealers. a, glass tube with lid; b, glass pearls inserted inside the tube; c, obturated bovine root within the glass pearls; d, addition of bile aesculin agar in the tube; e, *E. faecalis* ATCC 29212 onto agar culture medium; f, *E. faecalis* in broth culture medium; g, root inoculation; h, inoculated root; i, main possible results after incubation (i1, no bacterial microleakage; i2, visualization of contamination by external root pathway; i3, visualization of bacterial microleakage by obturation pathway) (Created with [BioRender.com/4bx5yxs](https://www.biorender.com/4bx5yxs)).

CAUTIONS

CAUTION: *E. faecalis* is a Biosafety Level 2 [BSL-2] pathogen. Follow all appropriate guidelines and regulations for the use and handling of pathogenic microorganisms.

CAUTION: The use of Personal Protective Equipment (PPE), like gloves, face shield and mask, during all protocols is essential to avoid contamination of the operator and the sample.

CAUTION: Microbiological steps should be performed in an aseptic environment such as a biosafety cabinet (according to the manufacturer's instructions) or working area of the bunsen burner to avoid contamination.

CAUTION: Dental procedures should be performed by a specialist in the field. It is important to highlight that procedures involving root canals must be performed by a specialist in endodontics.

CAUTION: Microbiological procedures must be performed by a microbiological specialist.

STRATEGIC PLANNING

Obtaining the number of samples of freshly extracted bovine teeth (basic protocol 1) must be carefully planned prior to the experiment. This will depend on previous contact with a slaughterhouse, so that they can plan and reserve the number of teeth requested.

BASIC PROTOCOL 1

Basic protocol title:

BOVINE TEETH SELECTION AND CLEANING

Introductory paragraph:

This protocol describes a method to select bovine teeth by visual inspection and then clean them using a periodontal curette. It is important to use PPE to protect the operator from contamination. This protocol allows the selection of clean bovine teeth with completely formed roots and root size of at least 22 mm.

Materials:

Fresh bovine teeth
0.9% sterile saline (see recipe)

Gloves
Ruler
Face shield
Periodontal Gracey curette (Golgran-Millennium, São Caetano do Sul, SP, Brazil)
Plastic flasks with lid
Refrigerator

Visual evaluation

1. Obtain fresh bovine teeth from a slaughterhouse after the animal has been slaughtered (a routine activity in the commercialization of meat for nutritional purposes).

The Brazilian Council for the Control of Animal Experimentation, as well as IACUC (Institutional Animal Care and Use Committee), only evaluate the use of live animals in research or animals that have been euthanized for research purposes, which was not the case in this protocol, so ethical approval is not applicable in this case.

2. Wear gloves to avoid contamination.

Fresh bovine teeth routinely come with biological materials, such as bone and gum debris, hence the need for protection with gloves when handling these teeth.

3. Perform visual inspection.

It is necessary to be in a well-lit environment to assure visual acuity.

4. Select bovine teeth with completely formed roots.

5. Select bovine teeth that have roots with at least 22 mm (Fig. 2).

6. Select bovine teeth without cracks in the roots.

Cleaning

7. Wear a face shield to protect the operator's face.

The face shield is necessary to protect the face against the detachment of bone fragments and calculus from the bovine teeth during the cleaning process.

8. Remove any external teeth structures, such as bone and calculus (Fig. 2a), using a periodontal Gracey curette (Golgran-Millennium).

9. Store clean teeth in 0.9% sterile saline solution in a plastic flask with lid in a refrigerator at 4 °C.

Replace the sterile saline solution once a week.

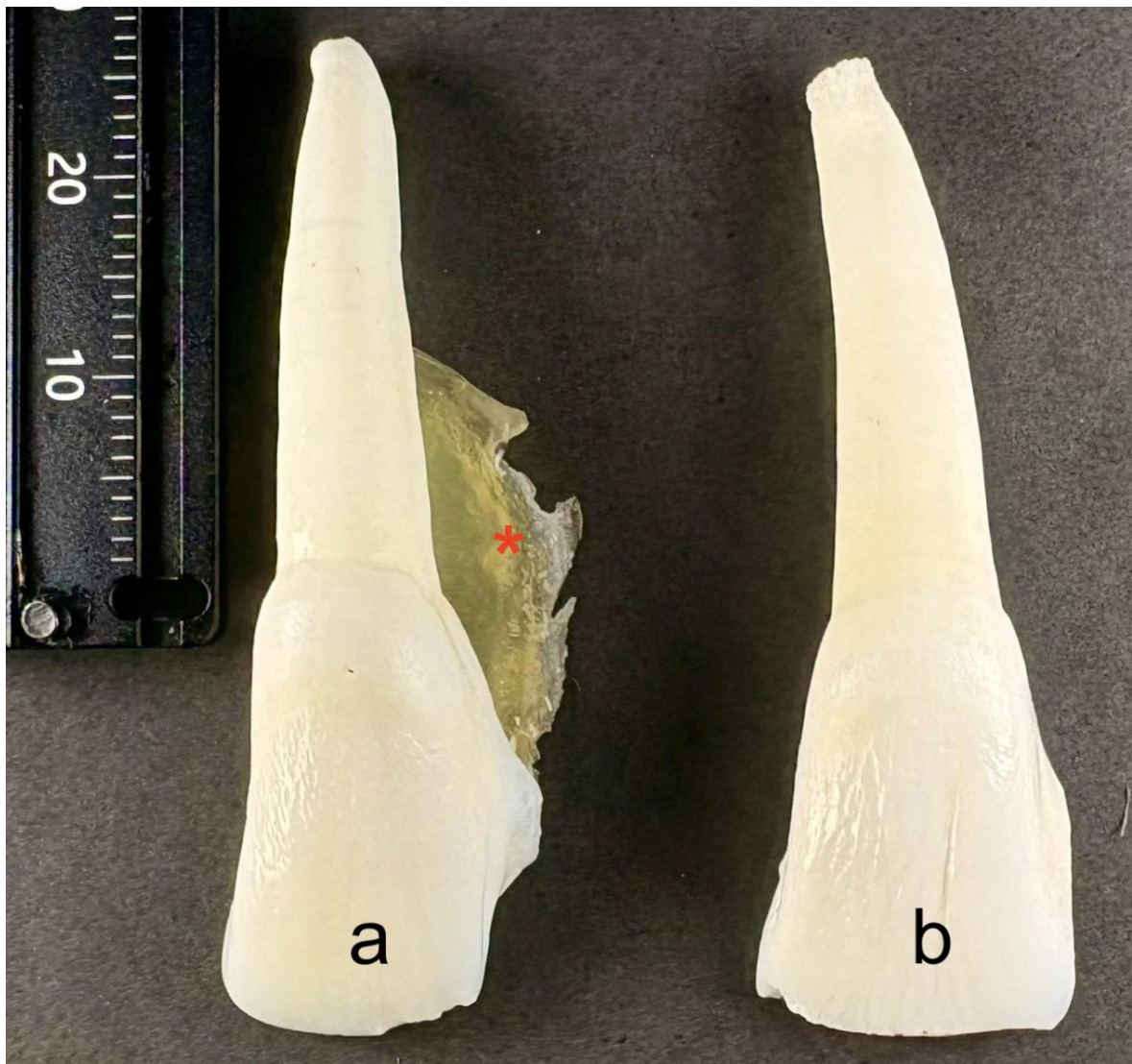


Figure 2. Selection of bovine teeth with fully formed roots and a minimum length of 22 mm. a, bovine tooth showing external structures, including an amorphous yellow structure (indicated by *). This external structure must be removed prior to use in the study. b, clean bovine tooth with no external structures.

BASIC PROTOCOL 2

Basic protocol title:

BOVINE TEETH INCLUSION

Introductory paragraph:

Basic protocol 2 describes a method to include teeth and store them. The radiographic evaluation with DIOX® intraoral portable X-ray (Micro Imagem Industry Commerce Import and Export LTDA, Indaiatuba, SP, Brazil), and exposure in digital phosphor plate systems with VistaScan Nano View (Durr Dental, Porto Alegre, RS, Brazil) allows the inclusion of bovine teeth with a single root canal from those initially selected in basic protocol 1. It is important to use personal protective equipment to protect the operator from secondary radiations.

Materials:

0.9% sterile saline solution

X-ray protective clothing

PPE

Wax Utility (Lysanda Dental Products Ltd, São Paulo, SP, Brazil)

#15 blade

Phosphor Plate digital systems (Durr Dental)

Selected teeth from basic protocol 1

DIOX® intraoral portable X-ray (Micro Imagem Industry Commerce Import and Export LTDA)

Digital systems (VistaScan Nano Easy, Porto Alegre, RS, Brazil)

Notebook

Software Vista Soft 3.0 (Durr Dental, Porto Alegre, RS, Brazil)

Flask with lid

Refrigerator

Radiographic evaluation

1. Wear X-ray protective clothing and PPE.

The use of radiographic protective clothing is necessary to avoid incidence of secondary radiation on the operator from the X-ray machine.

2. Cut 4 cm x 1 cm of the utility wax (Lysanda Dental Products Ltd) using #15 blade.

It is not necessary to use a handle for the scalpel blade because the wax is soft.

3. Position the utility wax (Lysanda Dental Products Ltd) on the long axis of the phosphor plate (Durr Dental) on a flat horizontal surface.

4. Place the tooth on the utility wax (Lysanda Dental Products Ltd) so that it is on its side on top of the phosphor plate (Durr Dental).

5. Sensitize the phosphor plate (Durr Dental).

Use the DIOX® (Micro Imaging Industry Commerce Import and Export LTDA) intraoral portable X-ray at a distance of 10 cm from the phosphor plate (Durr Dental), with 35 s exposure, to obtain the sensitization of the phosphor plate (Durr Dental).

6. Scan the image to produce a digital image.

Use the digital system (VistaScan Nano Easy) connected to a notebook, insert the phosphor plate (Durr Dental) in this system in order to produce a digital image in the Vista Soft 3.0 software (Durr Dental).

7. Exclude teeth with multiple root canals, lateral canals and apical deltas (Fig. 3a-b).

8. Include only bovine teeth with a single root canal (Fig. 3c).

Bovine teeth storage

9. Fill the flasks with 0.9% sterile saline solution.

10. Put the included teeth in the flasks and close the lids.

11. Store the flasks containing the included teeth in a refrigerator at 4 °C.

Replace the sterile saline solution once a week.



Figure 3. Digital image of bovine teeth. a, excluded tooth due to apical delta; b, excluded tooth due to lateral canals and apical delta; c, included tooth presenting a single root canal. Arrows: red arrows indicate apical delta and blue arrow indicates lateral canals.

BASIC PROTOCOL 3

Basic protocol title:

TOOTH STANDARDIZATION

Introductory paragraph:

In order to obtain a standardized and clean sample of bovine roots, basic protocol 3 describes the decoronation process as well as the standardization regarding shape and size of the root canals, the isolation of the external root face and

sterilization of the root samples. It is important to use PPE to avoid debris injuring the face during decoronation and root canal preparation.

Materials:

2.5% sodium hypochlorite (NaOCl)

17% ethylenediaminetetraacetic acid (EDTA) (Lysanda Dental Products Ltd)

PPE (Lab coat, Face shield, mask and gloves)

Double sided diamond cutting disc (Microdont, Socorro, SP, Brazil)

Mandrel clamping device (Microdont)

Dental Laboratory electric motor with suspension engine (Metalurgica FAVA Industria e Comercio, São Paulo, SP, Brazil)

Endodontic Ruler

Permanent marker

Bench vise

10 mL syringes with needle

K-file #10 (Dentsply Sirona, Pirassununga, SP, Brazil)

K-file #15 (Dentsply Sirona)

Largo drill #3 (Dentsply Sirona)

Largo drill #4 (Dentsply Sirona)

Largo drill #5 (Dentsply Sirona)

Electric motor with contra-angle VDW Silver Reciproc® (Dentsply Sirona)

W File #20.07 mechanized reciprocating files (TDK, Shenzhen Superline Technology Co, Pingshan, Shenzhen, GD, China)

W File #25.07 mechanized reciprocating files (TDK)

W File #35.06 mechanized reciprocating files (TDK)

W File #45.05 mechanized reciprocating files (TDK)

Utility Wax (Lysanda Dental Products Ltd)

Easy clean (Bassi Equipamentos, Belo Horizonte, MG, Brasil)

White nail varnish

Surgical grade paper

Surgical grade sealer

Autoclave

Bovine tooth decoronation

1. Wear PPE to avoid accidents and contamination.

It is important to use PPE (lab coat, face shield, mask and gloves) at this stage due to the formation of dentin dust during decoronation, which can lead to eye irritation and airway dryness.

2. Connect the double sided diamond (Microdont) cutting disc on the mandrel (Microdont) clamping device and attach them to the straight handpiece of the dental laboratory electric motor (Metalurgica FAVA Industria e Comercio).

3. Cut the crown of each tooth 2-3 mm below the cemento-enamel junction.

Standardization of the bovine root's cervical third

4. Standardize all roots with 22 mm.

Use an endodontic ruler and a permanent marker to trace a 22 mm distance from the apical region to the cervical region of each root.

5. Insert the bovine tooth in the bench vise and fill it with 10 mL of sodium hypochlorite 2.5%, using a syringe with needle.

6. Introduce the #10 file (Dentsply Sirona) into the root canal with oscillatory and passive cyclic motions (clockwise to fit the canal wall and counterclockwise to cut with controlled pressure).

During this process, irrigate the canal with 2.5% sodium hypochlorite using the 10 mL syringe with the needle attached, before and between each cycle to remove debris until #10 (Dentsply Sirona) file reaches the apex (working length).

7. Repeat exactly the same kinematics presented in step 6 with the #15 file.

8. Connect the #3 Largo (Dentsply Sirona) drill on the oscillating contra-angle of the VDW Silver Reciproc® electric motor(Dentsply Sirona).

9. Enlarge the cervical preflaring of the root with the #3 Largo drills (Dentsply Sirona) in continuous rotation at 1000 rpm.

Insert only the active part of the Largo drill(Dentsply Sirona) in the canal.

10. Repeat steps 8 and 9 with a #4 Largo drill(Dentsply Sirona)

11. Repeat steps 8 and 9 with a #5 Largo drill(Dentsply Sirona).

Standardization of the bovine root's third medium apex

12. Connect the #20.07 W file (TDK) on the oscillating contra-angle of the VDW Silver Reciproc® electric motor(Dentsply Sirona).

13. Gently, introduce the file into the root canal, advancing apically with a pecking motion (descending and ascending motion) in sets of three pecks. Repeat this step until you reach the apex.

During this process, irrigate the channel with 2.5% sodium hypochlorite before and between each set of three pecks to remove debris. After each set of three pecks, remove the instrument from the root canal and clean the file.

14. Repeat steps 12 and 13 with W File #25.07 (TDK) mechanized reciprocating files.

15. Repeat steps 12 and 13 with W File #35.06 (TDK) mechanized reciprocating files.

16. Repeat steps 12 and 13 with W File #45.05 (TDK) mechanized reciprocating files.

Cleansing of the bovine root canal

17. Close the apex of the root, temporarily, using the smallest possible amount of utility wax (Lysanda Dental Products Ltd).

This step allows the irrigating substances to remain in the canal during the cleansing process.

18. Fill a 10 mL syringe with 17% EDTA (Lysanda Dental Products Ltd) and another 10 mL syringe with 2.5% NaOCl.

19. Gently irrigate the bovine root canal with 2 mL of 17% EDTA (Lysanda Dental Products Ltd).

20. Use the Easy Clean instrument (Bassi Equipamentos), attached to the contra-angle of the VDW Silver ® reciprocating motor(Dentsply Sirona), in continuous rotation at 1000 rpm for 20 seconds in order to activate the EDTA (Lysanda Dental Products Ltd).

21. Gently irrigate the bovine root canal with 2 mL of 2.5% NaOCl.

22. Use the Easy Clean instrument, attached to the contra-angle of the VDW Silver ® reciprocating motor(Dentsply Sirona), in continuous rotation at 1000 rpm for 20 seconds in order to activate the NaOCl.

23. Repeat steps 19 through 22 two more times.

Steps 5 to 23 comprise the endodontic chemomechanical preparation.

Sealing and sterilizing the bovine teeth

24. After cleaning the bovine root canals, leave them at room temperature for 24h.

It is important to wait 24 hours so that the bovine rennet can dry completely.

25. Apply two coats of nail polish to all root surfaces except the apex and on the coronal plane. Wait 24 h between each application of nail polish.

This step allows us to isolate possible cracks and side channels on the external root that were not previously observed.

26. Twenty-four hours after drying the second layer of enamel, sterilize the bovine roots by autoclaving using a surgical grade pack for each root.

BASIC PROTOCOL 4

Basic protocol title:

SET PREPARATION

Introductory paragraph:

Basic protocol 4 describes the method for obtaining sterile sets composed of: glass pearls, a bovine root filled with the test endodontic sealer and solid culture medium in a glass tube with a lid. It is important to use aseptic techniques to avoid any contamination of the set. Although this protocol does not describe the process of obturation with endodontic sealers, we strongly recommend that this process be performed strictly following the manufacturer's instructions and using aseptic techniques in an aseptic environment.

Materials:

70% ethanol
Bile aesculin agar (see recipe), 55° to 60°C

Glass pearls (3 mm)
Analytical balance
5 mL borosilicate glass tube (16 x 41 mm) with lid
Surgical grade paper
Surgical grade sealer
Autoclave
100 mL borosilicate glass flask with lid
Water bath
Tweezers
Bunsen burner or biosafety cabinet
Sterile tips
Micropipette
PPE
Bacteriological incubator

1. Weigh 3.5 g of glass pearls
2. In a 5 mL borosilicate glass tube with lid (Fig. 4a), insert the 3.5 g of glass pearls (Fig. 4b).
3. Autoclave (121°C/20 min) these tubes with glass pearls.
4. After autoclaving the bile aesculin agar medium in the 100 mL borosilicate glass flask with lid, keep this medium in a water bath previously set to 60°C.

The water bath is important to keep the bile aesculin agar medium between 55° to 60°C, allowing the medium to remain in the liquid state after the autoclaving process.

5. Perform the root obturation with the endodontic sealer being tested, strictly following the manufacturer's instructions.

After obturation remove the excess endodontic sealer, using a # 4 largo drill, keeping 3 mm of obturation. The smear layer should be removed with 17% EDTA under agitation. Perform step 5 using aseptic techniques in an aseptic environment.

6. Insert aseptically, using sterilized tweezers, each root (immediately after root canal obturation with the test sealer) within the glass pearls 5 mm short of the bottom of the 5 mL glass tube (Fig 4c).

Make sure the tubes with glass pearls are completely dry before performing step 5.

7. Insert aseptically and gently, 1.6 mL (2 x 800 µL) of the bile aesculin agar medium (at 55 °C to 60 °C) in each 5 mL glass tube. This insertion needs to be performed with the sterile tip (previously attached in the micropipettor) leaning against the internal wall of the 5 mL tube to avoid touching the root (Fig 4d). Do not move the tube until the medium solidifies.

To avoid contamination, steps 5 and 6 need to be performed using aseptic techniques (wear PPE and disinfect gloves and surfaces with 70% ethanol) in an aseptic environment such as biosafety cabinet, laminar flow hood or working area of the bunsen burner.

8. Set the bacteriological incubator to 36 °C and incubate the sets (tubes containing root, glass pearls and medium) 24 h before inoculating them.

This 24h period is important to observe any contamination that might have occurred during set preparation as well as allow the hardening of endodontic sealers.

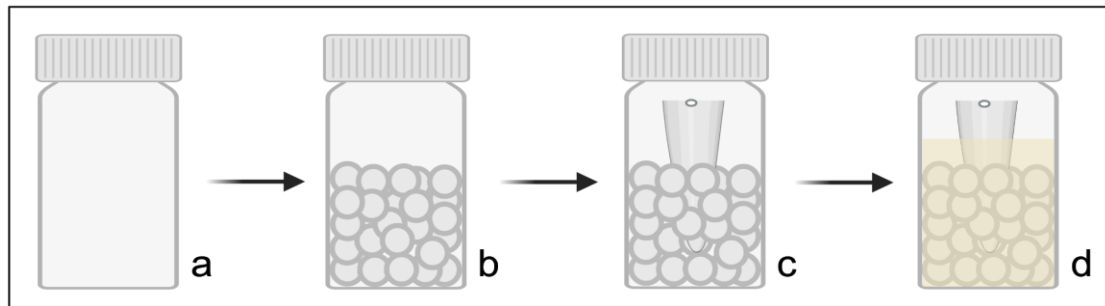


Figure 4. Set preparation. a, empty 5 mL borosilicate glass tube (16 x 41 mm) with lid; b, 3.5 g of glass pearls inside this tube; c, bovine root immediately after obturation with test sealer inserted the glass pearls 5 mm short of the bottom of the tube; d, full set composed of: the root, glass pearls, 1.6 mL of the bile aesculin agar medium inside the glass tube with lid. (Created with BioRender.com/1j7mipz).

BASIC PROTOCOL 5

Basic protocol title:

INOCULUM PREPARATION

Introductory paragraph:

This protocol describes how to obtain an inoculum with 10^9 Colony Forming Units (CFU)/mL of *E. faecalis* American Type Culture Collection (ATCC) 29212, from a frozen glycerol stock, in Brain Heart Infusion (BHI) broth. To avoid contamination rigorous aseptic techniques are necessary in all steps of basic protocol 5. For example, perform all steps wearing PPE in an aseptic environment such as a biosafety cabinet or working area of the bunsen burner, and disinfect gloves and surfaces with 70% ethanol.

Materials:

E. faecalis ATCC 29212 frozen stock in BHI broth with glycerol (at final concentration of 20% vol./vol. glycerol)
BHI agar (see recipe)
BHI broth (see recipe)
70% ethanol

PPE

Bunsen burner or biosafety cabinet
Sterile inoculation loop.
Bacteriological incubator
13 x 100 mm glass tubes
Spectrophotometer (Marca)

1. Using aseptic technique remove a small chunk of frozen *E. faecalis* from the cryotube, using a sterile inoculating loop.
2. Streak *E. faecalis* onto the BHI agar plate.

Starting from the edge of the Petri dish, in the first quadrant using the sterile inoculation loop streak the small chunk of *E. faecalis*, gently over one small section of the agar surface in a zigzag pattern. Using another sterile inoculation loop, drag some

bacteria from the last 3-5 lines of the zigzag pattern of the first quadrant into the second quadrant. Repeat this streak process for a third and fourth quadrant using new sterile inoculation loops for each quadrant and dragging from the previous area. This process dilutes the bacteria over the plate to isolate single bacterial colonies.

3. Incubate the plate at 36 °C for 24 h in an aerobic atmosphere to obtain isolated and pure colonies of *E. faecalis* (Fig. 5) (Fig. 6a).

Incubate the plate upside down (lid down).

4. Using aseptic technique and a sterile inoculation loop, transfer isolated and pure colonies to a 13 x 100 mm tube containing 5 mL sterile BHI broth (Fig. 6b) until reaching an optical density of 1.5 in 520 nm (equivalent to 10^9 CFU/mL).

It is important to strictly follow the instructions of the manufacturer of the tube spectrophotometer used.

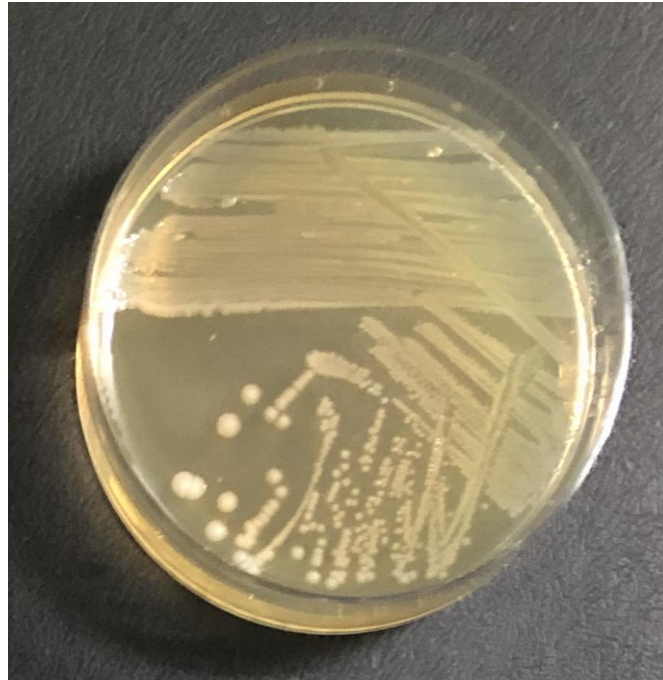


Figure 5. *E. faecalis* ATCC 29212 onto BHI agar after incubation at 36 °C for 24 h.

BASIC PROTOCOL 6

Basic protocol title:

CANAL INOCULATION

Introductory paragraph:

Basic protocol 6 provides inoculated root canals in the sets prepared in basic protocol 4. During the inoculation process PPE use, an aseptic environment (such as a biosafety cabinet or working area of the bunsen burner), and disinfecting of gloves and environment surfaces with 70% ethanol, are recommended.

Materials:

Inoculum (basic protocol 5)

PPE

Biosafety cabinet or bunsen burner

Sterile tips

Micropipette
 Sterile surface
 Sterile 1 mL syringe with needle
 Bacteriological incubator

1. Pipette 20 μ L of the inoculum with an aseptic micropipette.
2. Dispense the content on a sterile surface.
3. Aspire the inoculum using a sterile 1 mL syringe with needle.

It is important to avoid air aspiration in this step.

4. Carefully dispense the inoculum inside the root canal in the most apical region (Fig. 6c) and immediately close the set containing the inoculated root (Fig. 6d)

If using the working area of the bunsen burner, flame the neck of the 5 mL tube immediately after opening and before closing. This step provides a current that pushes airborne particles away from the opening, reducing the risk of microorganisms entering the tube and preventing contamination.

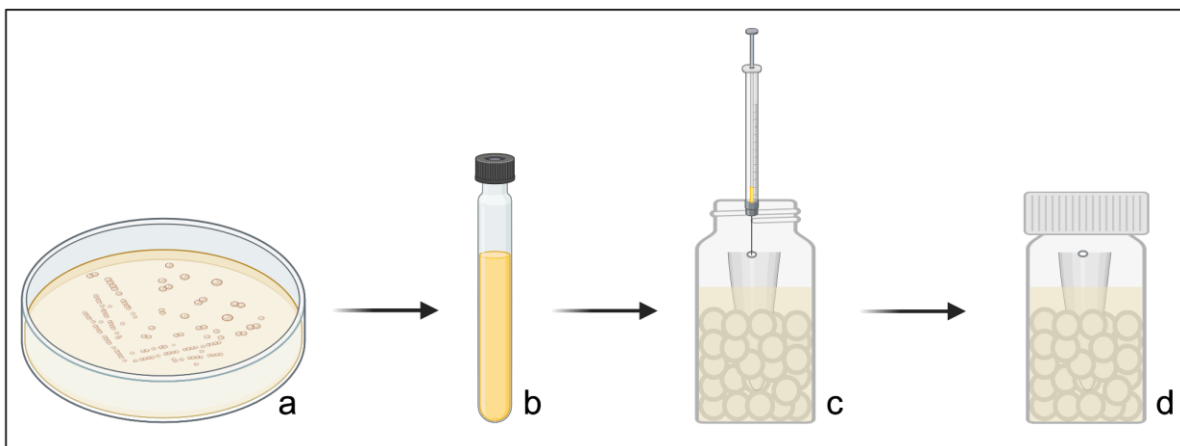


Figure 6. Root canal inoculation. a, *E. faecalis* ATCC 29212 grown on BHI agar after incubation at 36 °C for 24 h; b, inoculum of *E. faecalis* in BHI broth; c, inoculation process with sterile syringe with needle; d, set containing the inoculated root. (Created with BioRender.com/wr26vyl).

BASIC PROTOCOL 7

Basic protocol title:

SET INCUBATION

Introductory paragraph:

This protocol (basic protocol 7) describes how to incubate and maintain the sets throughout all 6 weeks of the experiment. During the incubation process it is necessary to add sterile fresh medium inside the root canal to guarantee the presence of nutrients and avoid drying out the medium. The use of PPE, an aseptic environment (such as biosafety cabinet or working area of the bunsen burner), and disinfection of gloves and surfaces with 70% ethanol are important to prevent contamination.

Materials:

70% ethanol
 Enterococcosel broth medium (see recipe)
 BHI broth (see recipe)

Bacteriological incubator
 PPE

Biosafety cabinet or bunsen burner
 Sterile tips
 Micropipette
 Sterile surface
 1 mL syringe with needle
 Sterile tweezers
 #30 sterile endodontic paper point (TDK)

13 x 100 mm glass tubes

1. Immediately after the inoculation, incubate the set at 36 °C for 6 weeks in an aerobic atmosphere.

To avoid leakage of the inoculum to the outside of the tooth, gently open and close the incubator door (this avoids shaking the incubated sets), and keep the sets vertically throughout the experiment. Using a rack adjusted to the diameter of the tubes can help keep them in the vertical position.

2. Two days (48 h) after inoculation dispense carefully 20 µL of a fresh BHI broth without inoculum inside the root canal in the most apical region.

Pipette 20 µL of BHI broth with an aseptic micropipette, dispense the content on a sterile surface and aspirate it using a 1 mL syringe with needle. It is important to avoid air aspiration in this step.

3. Twice a week dispense carefully 20 µL of fresh sterile BHI broth inside the root canal in the most apical region.

Steps 2 and 3 ensure the viability of E. faecalis.

4. Observe all sets daily during 6 weeks to determine if and when bacterial microleakage occurred and by what pathway (Fig 7).

No bacterial microleakage results in no color change (Fig. 7a). A change from the bottom indicates leakage through the obturation (Fig 7b); from the top, it demonstrates surface contamination, and the set should be discarded (Fig 7c).

5. After 6 weeks of incubation, using a sterile tweezer, insert a #30 sterile paper point (TDK) for 1 min inside the root canals of all sets without bacterial microleakage (Fig. 7a).

6. Transfer the paper point (TDK), using a sterile tweezer, to a 13 x 100 mm tube with 1mL enterococcosel broth.

7. Incube the tube at 36 °C for 24 h in an aerobic atmosphere.

Steps 5-7 are important to confirm the presence of viable E. faecalis inside the root canals of all sets without color changes (Fig. 7a). Tubes without color change (Fig. 8a) indicate the absence of E. faecalis in the paper point. Whereas tubes with color change (Fig. 8b) indicate the presence of E. faecalis in the paper point. These steps guarantee that the unchanged set color (Fig. 7a) faithfully represents an absence of bacterial microleakage by obturation even with viable E. faecalis inside the root canals.

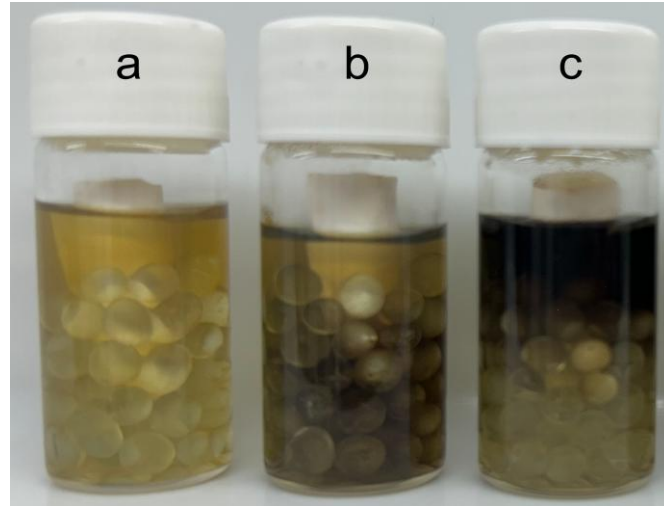


Figure 7. Sets representing the most common results that can be observed. a, control set (without inoculum) (unchanged color of the bile aesculin agar); b, microleakage by canal pathway (change of the color of bile aesculin agar to dark brown initiated from within the medium); c, surface contamination (change in color of bile aesculin agar to dark brown, beginning at the surface of the medium).

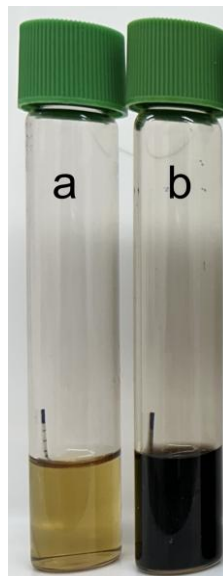


Figure 8. Tubes of enterococcosel broth containing paper points after 24 h incubation. a, the medium color remained unchanged (light brown/yellow) due the paper point inserted contained no *E. faecalis*; b, the medium color changed to black due to aesculin hydrolysis, resulting from the presence of *E. faecalis* in the broth.

REAGENTS AND SOLUTIONS:

Use distilled water to prepare all recipes. Sterilize all recipes by autoclaving for 15 min at 121 °C.

0.9% (w/v) sterile saline

- 0.9 g sodium chloride
- 100 mL distilled water
- Dissolve 0.9 g sodium chloride to each 100 mL distilled water
- Autoclave to sterilize and store ≤6 months at 4°C

Bile aesculin agar medium (Neogen, cat. no. NCM0117A)

- 3.68 g bile esculin agar medium powder
- 80 mL distilled water
- Dissolve 3.68 g bile esculin agar medium powder for each 80 mL distilled water.
- Stir while heating until all solids are dissolved.
- Autoclave to sterilize 15 min at 121 °C

If the medium is not going to be used immediately, let it solidify at room temperature and store ≤3 month at 4°C. Melt it in a microwave or in a boiling water bath prior to use.

BHI agar medium (Becton, Dickinson and Company, cat. no. BD241830)

- 52 g BHI agar medium powder
- 1000 mL distilled water
- Add 52 g BHI agar medium powder and ultrapure water to a final volume of 1000 mL in a 2 L flask
- Stir while heating until all solids are dissolved
- Autoclave to sterilize 15 min at 121 °C
- Pour 12 mL into sterile 60 mm × 15 mm Petri dishes
- Allow to solidify at room temperature
- Incube the plates at 36 °C to 24 h to observe any contamination and exclude contaminated plates, if there are any
- Store ≤3 months at 4°C in sealed bags at 4°C

Wait for the BHI plates to cool down prior to inoculation with bacteria.

BHI broth medium (Becton, Dickinson and Company, cat. no. BD237500)

- 3.7 g BHI broth medium powder
- 100 mL distilled water
- Combine in a 200 mL flask
- Stir until all solids are dissolved
- Aliquot 5 mL into a 13 x 100 mm tube with lid
- Autoclave for 15 min at 121 °C
- Store ≤6 months at 4°C

Ensure that BHI broth medium is sterile before use (sterile BHI broth medium is clear and not cloudy). Wait for the medium to cool down prior to use.

Enterococcosel broth medium (Becton, Dickinson and Company, cat. no. BD212207)

- 4.3 g bile enterococcosel broth medium powder
- 100 mL distilled water

- Combine in a 200 mL flask
- Stir until all solids are dissolved
- Aliquot 1 mL into a 13 x 100 mm tube with lid
- Autoclave for 15 min at 121 °C
- Store ≤6 months at 4°C

Ensure that bile enterococcosel broth medium is sterile before use (sterile enterococcosel broth medium is clear and not cloudy). Wait for the medium to cool down prior to use.

COMMENTARY:

Critical Parameters:

Basic protocol 1:

When working with fresh bovine teeth, it is worth considering the anatomical diversity of the teeth, making it necessary to have a larger quantity due to the high exclusion rate. It is important to store them in flasks with lids containing 0.9% sterile saline. To keep the teeth hydrated, with their characteristics, and prevent dehydration, maintain the flasks refrigerated at 4°C. The use of a magnifying lens and incident light can improve the visualisation during selection of bovine teeth with completely formed roots and without visual cracks in the roots. It's important to sharpen the Gracey periodontal curette before removing external teeth structures (e.g: bone, calculus). This protocol should be performed by a dentist or a dental hygienist, depending on the local legislation.

Basic protocol 2:

It is important that the digital phosphor plate is intact and that the intraoral portable X-ray and the notebook batteries are charged. During X-rays of the bovine teeth procedure, it is important to fix the teeth in the utility wax to prevent displacement, avoiding distortions in the radiographic images. This protocol must be performed by a trained operator.

Basic protocol 3:

For effective decoronation of a bovine tooth, always use a double sided diamond cutting disc. It is important to screw the disc completely into the mandrel to avoid it from coming loose during use. Once the disc comes loose, it becomes useless. Keeping the teeth hydrated during the cutting process prevents excess dust. After removing the crown, return the root to the refrigerated saline solution.

During canal preparation, secure the bovine root in a bench vise to prevent displacement and charge endodontic electric motor battery, allowing the operator to perform root canal preparation. Use the endodontic files according to manufacturer instructions to avoid their fracture during preparation.

It is important to dry the outer surface of the root to close the apex with utility wax before cleansing of the root canal systems.

To allow the operator to seal the outer surface of the root with nail polish, the use of an object such as a toothpick into the root canal may be helpful. This object can be inserted into a surface like styrofoam until it's completely dry. To avoid possible adhesion of nail polish to the plastic part of the surgical paper, do not use a tight surgical paper package.

It is important that procedures involving root canal be performed by a specialist in endodontics.

Basic protocol 4:

Before inserting glass pearls into the tubes, check and discard any glass pearls with a color other than clear and make sure they are clean and dry, dark pearls may lead to false positive interpretation. A funnel-like device can be used to assist in inserting the correct amount of pearls into the tube.

Immediately after root canal filling with the test sealer, insert it into the tube within the glass pearls and fill with agar medium to mimic the moist environment of the oral cavity. In this step it is important to maintain the bile aesculin agar in liquid form, in a water bath between 55° to 60°C, prior to inserting it into the tube. To carry this out, before placing the

flask with agar medium in the water bath, it must be completely melted. To do this, use it immediately after autoclaving or, if it is stored at 4°C, melt it in a microwave or in a boiling water bath.

Basic protocol 5:

Before streaking *E. faecalis* onto BHI agar plates, ensure the plates with medium are sterile. To do so, incubate them at 36°C for up to 24 h after preparation to monitor contamination. To cool down the agar plates before inoculating with *E. faecalis*, place them in an incubator after removing them from the refrigerator.

Before transferring colonies to a tube with BHI broth to prepare the inoculum, ensure the broth is sterile, incubate it at 36°C for up to 24 h after preparation to monitor contamination.

Ensure that the *E. faecalis* on the BHI agar plate is pure. To achieve this, it is important to streak *E. faecalis* to obtain isolated bacterial colonies. Incubate the plate upside down (lid down) and, after the incubation period, carefully observe the morphological characteristics of the colonies. A stereoscopic microscope can aid this observation.

Protocol 5 must be performed by a microbiological trained operator, using aseptic technique and sterile tools throughout this protocol.

Basic protocol 6:

During inoculation, avoid allowing the inoculum to reach the surface taking the following precautions: i) when aspirating the inoculum with the syringe needle, avoid aspirating air (bubbles can cause the inoculum to leak, during inoculation process, on the bile aesculin agar surface); ii) carefully insert the needle into the canal, ensuring that the inoculum is gently dispensed only into this space. iii) when removing the needle from the canal, do not rest it in its walls to avoid projecting particles containing *E. faecalis* directly on the bile aesculin agar surface, iii) keep the sets vertically throughout the all experiment. When the inoculum reaches the surface of bile aesculin agar. When the color changes, starting from the surface, it indicates contamination (Fig. 7c).

A microbiological trained operator must perform this protocol, using aseptic technique and sterile tools during all steps.

Basic protocol 7:

During incubation, avoid allowing *E. faecalis* to reach the surface taking the following precautions: i) the bacteriological incubator door should open and close smoothly, avoiding vibrations and spillage of the inoculum; ii) when adding a new fresh medium inside the root canal, aspirate the BHI, avoid aspirating air, ensuring that the medium is gently dispensed only into the root canal, and do not rest the needle in the root walls when removing it from the canal; and iii) keep the sets vertically throughout the experiment, avoiding leaks of the bacteria onto the surface of the bile aesculin agar.

When the color changes, starting from the surface, it indicates contamination (Fig. 7c). Protocol 7 must be performed using sterile tools and aseptic technique by a microbiological trained operator.

Troubleshooting Table:

Common problems with the protocols, their causes, and potential solutions are presented in Table 1.

Table 1. Troubleshooting Guide for Evaluating Bacterial Microleakage of Endodontic Sealers

Problem	Possible Cause	Solution
High rate of teeth exclusion	Anatomical diversity of the teeth	Obtain a larger quantity than necessary
Tooth dehydration	The teeth were not kept hydrated	Keep the teeth in a saline solution inside flasks with lids, and maintain them refrigerated at 4°C
Difficulty to remove external teeth structures	Blunt Gracey periodontal curette	Sharpen the Gracey periodontal curette on the sharpening stone
Radiographic image showing risks,	Scratched or heavily used digital phosphor plate	Change the digital phosphor plate for a new one

alterations and/or low quality		
Disc unusable after little use	Insufficient screwing of the disc into the mandrel	Screw the disc completely into the mandrel to avoid it from coming loose during use
Utility wax does not stick to teeth	The bovine tooth is wet	Dry the tooth
Contamination of culture medium	Contamination during pouring the culture medium into the petri dish and/or during the microbiological process	Perform microbiological steps using aseptic techniques (wear PPE and disinfect gloves and environment surfaces with 70% ethanol) in an aseptic environment such as biosafety cabinet or working area of the bunsen burner
Set contamination	Leakage of the medium, from the inside of the canal, containing <i>E. faecalis</i> , and/or inoculation outside of the root canal	Avoid bubbles inside syringe containing inoculum and BHI broth; dispense inoculum or BHI broth only into the canal; avoid projecting particles containing <i>E. faecalis</i> ; open and close smoothly the bacteriological incubator; and keep the sets vertically during the length of the experiment. Train the operator before performing microbiological steps

Understanding Results:

Through visual selection and mechanical cleaning with a periodontal curette (Fig. 2), basic protocol 1 selects a homogeneous group of clean bovine teeth. These selected teeth must not have external structures (e.g: bone, calculus), nor cracks. Teeth with fully formed roots with a minimum length of 22 mm are essential for the execution of subsequent protocols. Proper storage in sterile saline solution at 4°C ensures the preservation of the structural integrity of the specimens until use.

Basic protocol 2, through digital radiographic evaluation, allows the identification of the anatomical characteristics of the root canal. Teeth with multiple canals, lateral canals, and apical deltas should be excluded (Fig. 3a-b), and included teeth that present a single root canal (Fig. 3c), enabling the use of similar specimens in this proposed protocol. The decoronation process, chemomechanical endodontic preparation up to #45.06, external sealing of the roots with nail polish and autoclaving, results in a group of sterile 22 mm bovine roots with isolated external faces and their single canals are standardized and cleaned at the end of basic protocol 3.

Basic protocol 4 presents a preparation of sterile sets to be used in this proposed protocol. Each set is composed of a borosilicate glass tube containing glass pearls, obturated bovine root and bile aesculin agar (Fig. 4). This basic protocol also describes how to detect possible contamination during set preparation. Contaminated sets should be discarded.

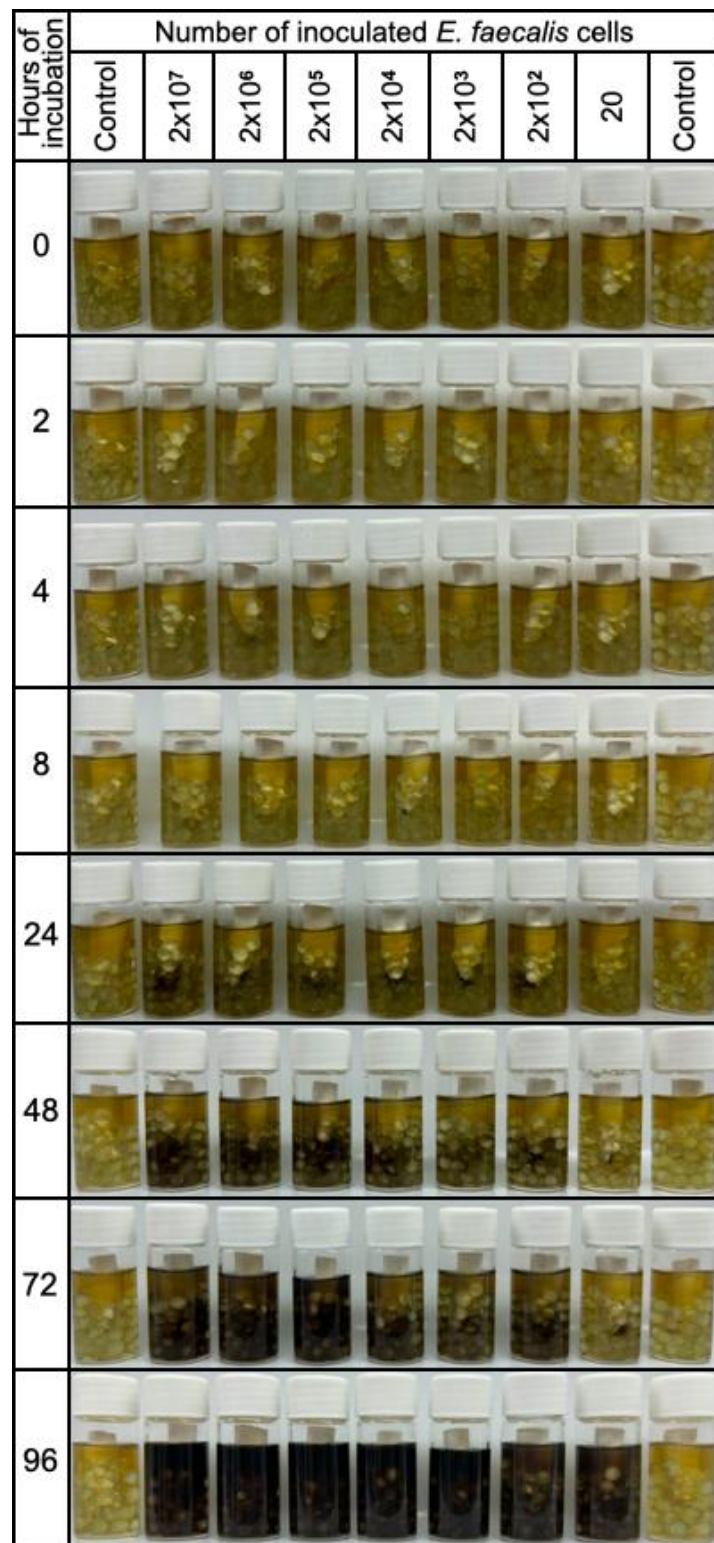
Basic protocol 5 describes the revival of *E. faecalis* ATCC 2921 from a frozen stock culture. After streaking and incubation of 36 °C for 24 h, isolated colonies should be visible away from the initial streaking area (Fig. 5). Additionally, the preparation of an inoculum from isolated and pure *E. faecalis* colonies (Fig. 6a-b) containing 10⁹ CFU per 1 mL of BHI broth is presented in this protocol.

The methods described in basic protocols 6 and 7 allow us to obtain inoculated sets (Fig. 6c) with 20 µL (~2x10⁷ CFU/mL) of *E. faecalis* inside the root canal, incubated vertically for 6 weeks, with addition of fresh medium inside the root canals twice a week, and daily observed to detect changes of color of the bile aesculin agar (Fig. 7). Figure 7 represents the most common results that can be observed in this protocol, being: i) unchanged color of the medium (bile aesculin agar), resulting from the absence of bacterial microleakage by the endodontic sealer used to obturate the root canals (Fig. 7a); ii) change

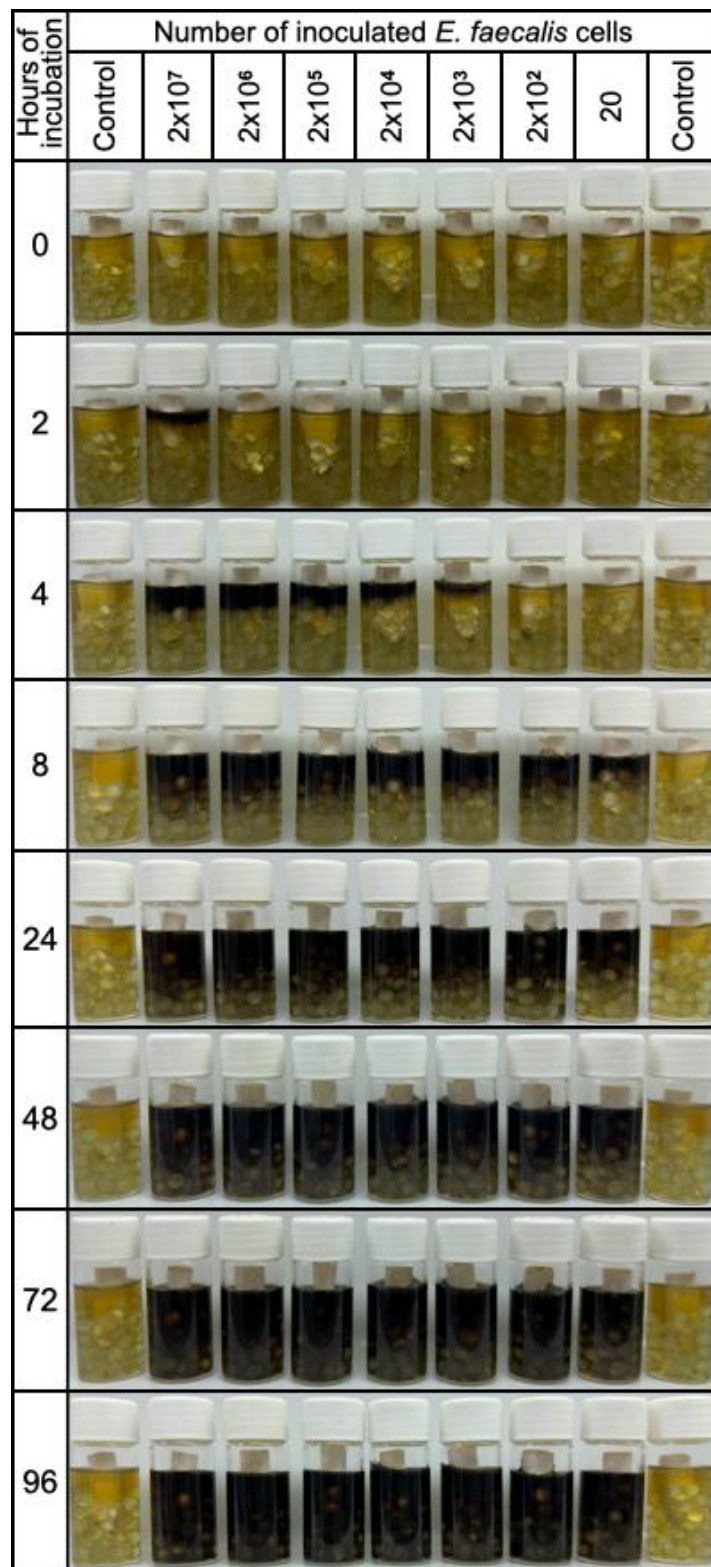
of the color of bile aesculin agar from yellow/light brown to dark brown initiated from within the medium, due to the hydrolysis of aesculin by *E. faecalis*, resulting from the presence of this bacteria microleakage through the endodontic sealer used to obturate the root canals (Fig. 7b); iii) change in color of bile aesculin agar from yellow/light brown to dark brown, beginning at the surface of the medium, due to aesculin hydrolysis by *E. faecalis*, resulting from the presence of this bacterium on the agar surface, indicating contamination via the external root pathway; sets presenting this characteristics should be discarded (Fig. 7c).

At the end of the assay (6 weeks), sets such as the one shown in figure 7a need to confirm the presence of viable *E. faecalis* in the root canal. To do this, a sterile paper point can be inserted into the root canal and transferred to a broth that indicates presence of *E. faecalis*. After the incubation period, the esculetin, resulting from the hydrolysis of aesculin, reacts with ferric ions present in the medium, such as enterococcosel broth, forming a dark complex (Fig. 8b), while in the absence of the microorganisms that caused this reaction, the medium maintains its original color (Fig. 8a).

Figures 9 and 10 show different concentrations of *E. faecalis* over time inoculated either inside the root canal without obturation or externally to the root, respectively. The concentration of the inoculum is inversely proportional to the time required for the medium to change color, regardless of where the inoculation occurs. Comparing these figures, there is a clear difference when *E. faecalis* reaches the culture medium through the root canal (Fig. 9 - color is first seen and disseminates from the bottom upward) or through contamination on its surface (Fig.10 - color is first seen at the top and spreads down), regardless of the inoculum concentration. So, the tubes contaminated on the surface can be easily identified during daily visual inspection.



Figures 9: Different concentrations of *E. faecalis* inoculated inside the root canal over time.



Figures 10: Different concentrations of *E. faecalis* inoculated externally to the root (on the surface of the medium) over time.

Time Considerations:

The entire protocol can be completed in approximately 8 to 9 weeks, but depending on slaughterhouse availability as well as of the number of roots tested, the period can be considerably extended.

The selection and cleaning of bovine teeth (basic protocol 1) takes 1 day. Basic protocol 2 takes 1 day to take and analyze the radiographs, applying the inclusion criteria. The preparation of the teeth included in the study (basic protocol 3) is the most laborious phase of teeth preparation (basic protocol 1-3). It begins with the decoration of the bovine teeth, which takes 1 to 2 days, while the standardization of the cervical, middle, and apical thirds of the teeth can be performed in 1 to 2 days. Root canal cleaning takes 1 day, but an additional day (for complete drying) is required before performing the next step. Sealing the tooth with enamel takes 2 days. Basic protocol 4 needs about 2-3 days to prepare the sets containing glass pearls, bovine root with its canal filled with endodontic sealer, requiring an extra day before inoculating them. Inoculum preparation (basic protocol 5) and canal inoculation (basic protocol 6), require 2 days to prepare the media, streak the *E. faecalis*, prepare the inoculum and perform the inoculation. Basic protocol 7 (set incubation) requires daily observation for 6 weeks, with addition of fresh medium inside root canals twice a week.

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CONFLICT OF INTEREST STATEMENT:

All declare no conflicts of interest.

DATA AVAILABILITY STATEMENT:

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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4 CONSIDERAÇÕES GERAIS

Diante do exposto, conclui-se que a microinfiltração endodôntica representa um desafio persistente à previsibilidade e longevidade dos tratamentos endodônticos. A complexidade anatômica do sistema de canais radiculares, aliada à resistência microbiana, especialmente de biofilmes formados por *Enterococcus faecalis* e *Candida albicans*, compromete a eficácia da limpeza e do selamento do SCR. Embora os cimentos endodônticos atuais reúnam propriedades físico-químicas e biológicas desejáveis, sua ação ainda demanda avaliações contínuas por meio de modelos experimentais, sendo assim a adoção de protocolos laboratoriais, como os testes de microinfiltração microbiana, contribuem para o aprimoramento da avaliação dos materiais obturadores.

O avanço na compreensão da microinfiltração microbiana e na seleção de materiais obturadores, a partir dos ensaios laboratoriais, com maior capacidade de vedação é fundamental para reduzir a taxa de insucesso terapêutico e promover a excelência na prática endodôntica.

5 CONCLUSÃO

O protocolo desenvolvido neste estudo para verificar a microinfiltração microbiana apresentou as características desejadas para avaliar o selamento de cimentos endodônticos, uma vez que permite conhecer a via da microinfiltração.

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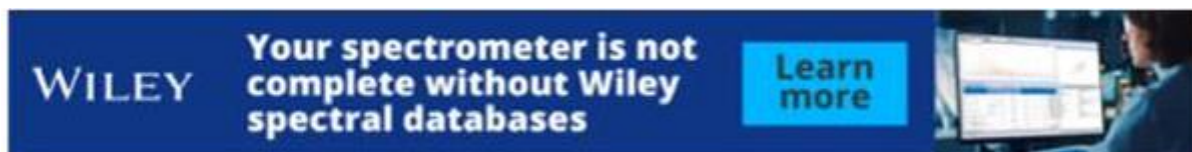
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APÊNDICE A — Normas para submissão de manuscritos ao periódico current protocols



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Peer Review

This journal operates under a single-anonymous [peer review model](#). For thirty years, Current Protocols has set the standard for excellence in laboratory methods in the biological sciences. Every article undergoes a rigorous, three-tiered review process that has been carefully optimized for protocol-based manuscripts. The process begins with our distinguished academic [editorial](#)

board members, who carefully consider potential new content and the best authors prior to invitation. Unsolicited manuscripts or proposals for manuscripts are also subject to this vetting process. Every manuscript is subject to expert peer review by members of our academic editorial board. Guest editors or additional reviewers may be called upon if needed, but experience in evaluating laboratory protocols is essential. The next layer of review is performed by a Wiley **in-house editor** with an advanced degree in a relevant subject. Representing the perspective of the audience – i.e., a novice to the technique – this editor is trained to anticipate and address the questions that may arise when performing a protocol. This in-house editor also considers thoroughness, presentation, clarity, and consistency with respect to the rest of the protocol collection. Based on these reviews, authors may be asked to revise a protocol prior to acceptance. The third and final revisions are done by scientific copyeditor, also a scientist with laboratory experience, then reviews the accepted manuscript in fine detail, requesting missing information, resolving any inconsistencies among protocol steps, and ensuring that every portion of the manuscript is complete so that each protocol works the first time, every time. Together, our editorial team works in close collaboration with the author to ensure that the final protocol is of the highest quality.

Manuscripts authored by editorial board members, will be sent to editors unaffiliated with the author or institution and monitored carefully to ensure there is no peer review bias.

Wiley's policy on the confidentiality of the review process is [available here](#).

Guidelines on Publishing and Research Ethics in Journal Articles

Current Protocols uses iThenticate's CrossCheck software to detect instances of overlapping and similar text in submitted manuscripts. Read [Wiley's Top 10 Publishing Ethics Tips for Authors](#) and [Wiley's Publication Ethics Guidelines](#).

Current Protocols follows the core practices of the [Committee on Publication Ethics \(COPE\)](#) and handles cases of research and publication misconduct accordingly (<https://publicationethics.org/core-practices>).

Conflict of interest (Col): Authors are responsible for disclosing all financial and personal relationships between themselves and others that might bias their work. Authors should describe the role of sponsor(s), if any, in protocol design and in the collection, analysis and interpretation of data. If the supporting source had no such involvement, the authors should so state. Cols must be clearly indicated in the manuscript accompanying the submission.

Human and animal subjects: If submitted protocols use animal or human subjects, authors must include a note in the protocol introduction that all experiments must be approved by the appropriate institutional and or national review boards and, in addition, that human subjects must give informed consent.

Guidelines on Biosecurity Issues: *Current Protocols* is dedicated to the responsible publishing of protocols to further advance scientific knowledge and its application for the benefit of human life.

Acknowledging that some experiments and microorganisms can be considered to have "dual use"—i.e., in which technology can be used legitimately for human betterment or misused for bioterrorism—the Committee on Research Standards and Practices to Prevent the Destructive Application of Biotechnology (headed by Gerald Fink and part of the National Research Council, Division on Policy and Global Affairs) has identified seven classes of experiments that warrant special review by experts. Based on these recommendations, it is the policy of the editorial boards of *Current Protocols* NOT to publish protocols that describe in detail experiments that:

- render human or animal vaccines ineffective;
- confer resistance to therapeutically useful antibiotics or antiviral agents for humans, animals, or crops;
- enhance the virulence of human, animal, or plant pathogens, or make nonpathogens virulent;
- increase the transmissibility of pathogens;
- alter the host range of pathogens;
- enable the evasion of diagnostic or detection methods; or
- enable the weaponization of biological agents or toxins.

Please review Wiley's policies surrounding human studies, animal studies, clinical trial registration, biosecurity, and research reporting guidelines.

Author Contributions

For all articles, the journal mandates the CRediT (Contribution Roles Taxonomy)—more information is available on our [Author Services](#) site. You will have the opportunity to select the appropriate CRediT contribution for each author when submitting your manuscript on the submission site.

Article Types and Manuscript Preparation

Article Types

Current Protocols does not impose length requirements on manuscripts (with the exception of Protocol Communications). The length of the manuscript should be dictated by the topic presented.

Authors should keep in mind that many readers of *Current Protocols*, while trained in the subject covered, are not experienced in a large proportion of the procedures described. Therefore, sufficient detail must be provided to permit duplication of the protocols in any laboratory, whatever

the disciplinary background or level of sophistication, for the benefit of the novice experimenter. It is preferable to provide too much detail that can be edited if needed, rather than not enough detail. The primary audience for our bioinformatics protocols consists of molecular biologists who have an interest in applying bioinformatics tools to their own work, but who may have little computational experience.

Protocol Article

The Protocol Article is the most common article type published in *Current Protocols*; unless otherwise noted in your invitation, please prepare your manuscript as a Protocol Article. The main feature is the step-by-step numbered set of detailed instructions, accompanied by an in-depth Commentary section describing background information, troubleshooting, and more, as detailed in the template. The level of detail should be such that a researcher new to the technique could perform the experiment successfully and confidently.

Download the Protocol Article Word template [here](#).

Clinical Protocols

Clinical protocols are articles describing a detailed account of the hypothesis, rationale, and methodology of a proposed or ongoing piece of prospective clinical research. Current Protocols is happy to accept clinical protocols for Basic experimental studies in humans (BESH), Phase 1-4 clinical trials and prospective cohort studies. Articles will generally only be considered for ongoing or future studies i.e. the clinical protocol should be submitted before data collection has ended. Please do not include pilot, baseline or preliminary data in clinical protocol articles. Current protocols will, however, accept protocols submitted after study completion and including data for innovative experimental approaches employed during trials such as novel diagnostic techniques; these should be submitted to a different section as a standard Current protocols article.

Download the Clinical Methods Word template [here](#).

Those submitting a Diagnostic Methods protocol can use our usual Word template [here](#).

All submitted protocols must have ethical approval (please include the name of the approving body in the methods), patient consent and interventional studies must be prospectively registered in a registry which aligns with the WHO's minimum standards for complete registration (WHO data set). The IRB approved clinical protocol, registration and submitted clinical protocol should match. Statistical Analysis Plans (SAPs) should be incorporated into the main body of the text.

Further to the above and Current Protocols standard policies authors should, when submitting, include:

- An update on the status of the research in the cover letter.
- A copy of the IRB approved clinical protocol (this must be in English).

- The registration number in the abstract. · A completed SPIRIT checklist (The SPIRIT Statement – GUIDANCE FOR CLINICAL TRIAL PROTOCOLS) for interventional trials or another appropriate reporting guideline checklist.
- A declaration, in the methods, of who will have access to the final trial dataset and if this is restricted, why.
- A declaration on the role of the funding body in the design, execution and analysis of the study.

Overview Article

An overview article is presented as explanatory text with no protocol steps. It is not meant to be a thorough literature review of a subject, but rather an introduction to the major concepts, strategies, and practical advice. Please do not format your manuscript as an overview unless invited to do so by the Editor.

Authors should bear in mind that many readers will be new to the subject matter and will be looking to the overview for a brief and accessible introduction to the topic of interest with a brief, targeted reference list that will help guide them to additional background in the literature. The reference list should generally contain a maximum of 25 to 30 references, of which the majority should be less than 5 years old and contain, when available, some recent review articles.

The format is quite flexible, but an Overview Article Word template is available [here](#).

Updated Protocol

Protocols evolve over time. We encourage authors of Protocol Articles to submit updates to their original article, detailing any changes made to the original protocol, alternative approaches, new reagents, new applications, etc. Please contact the Developmental Editor, who will provide a Word file of the published article for you to revise.

Updated Overview

We encourage authors of Overview Articles to submit updates to their original article, detailing new approaches, considerations and applications. Please contact the Developmental Editor, who will provide a Word file of the published article for you to revise.

Essential Laboratory Technique

Essential Laboratory Technique articles start with an introduction of the principles behind the experimental method, followed by the detailed protocols. These articles are written with a focus on training novice researchers who have not had practical experience outside of introductory laboratory courses. This article will serve as the reader's first experience with the techniques described.

Download the Essential Laboratory Technique Word template [here](#).

Protocol Communication

Protocol Communications are designed to enhance the replicability of findings reported in journal articles, by providing detailed instructions and exact conditions used to generate the data. Links will be provided between the Protocol Communication and the related journal article.

Protocol Communications are limited to 4000 words, 3 figures, 2 tables, and no more than 3 protocols per article.

Download the Protocol Communications Word template [here](#).

Manuscript Preparation

The value of *Current Protocols* to the scientific community depends upon the quality and comprehensiveness of the manuscripts. Every research question is unique; thus, the reader must be given enough information to tailor the protocol to their own needs. Author adherence to our style and manuscript templates is vital to the success of this endeavor.

The Editorials will work with you to define the scope and content of your manuscript. Please do not hesitate to discuss the content of your manuscript, or to make suggestions for additional content with the Editorial Board or the Developmental Editor.

Methods used in previous publications

Current Protocols' mission is to provide the most comprehensive collection of robust and reliable protocols as a resource for scientists at the bench. As such, *Current Protocols* does not publish novel, primary research or unproven methods. Our editors seek to include detailed protocols that are known to work and have been used previously to generate research findings. Thus, authors should be assured that submitting a protocol that has been part of a previously published article to *Current Protocols* is not problematic, and is, in fact, preferred. Even in instances where the basic steps have been included in a previous publication, self-plagiarism is usually not an issue. *Current Protocols* requires that manuscripts include rich detail and annotations at a level typically not found in "recipes." Furthermore, such recipes are simple instructions and are not subject to [copyright](#). Of

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Title Page

The title page should contain:

- i. A brief informative title containing the major key words. (see [Wiley's best practice SEO tips](#))
- ii. The full names of the authors
- iii. The author's institutional affiliations where the work was conducted, with a footnote for the author's present address if different from where the work was conducted; denote corresponding author and provide address, phone numbers, email address.

Main Text File

The main text file should be in Word and include:

- **Abstract:**

The abstract should be no more than 300 words in length and provide a summary of the contents of the manuscript. **For protocol articles**, please list the title of each protocol included in your manuscript beneath the abstract paragraph (shown below) – this text counts towards the word limit of your abstract.

Basic Protocol 1: (Title)

Basic Protocol 2: (Title) etc.

- **Keywords:** List 3-5 keywords appropriate to your article. Also include the same keywords in the Research Exchange upload
- **Main body:** You *must* use the appropriate template to prepare your manuscript. Please choose from one of the following templates:

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Essential Laboratory Techniques: [CP Essential Technique Template](#)

Protocol Communication: [CP Protocol Communication Template](#)

Overviews: [CP Overview Template](#)

- Conflict of Interest Statement (even if to state that no conflict exists)
- Acknowledgements (optional; this should include funding information if applicable)
- Data Availability Statement
- Literature Cited
- Tables (each table complete with title and footnotes)
- Figures: Figure legends must be added beneath each individual image during upload AND as a complete list in the text.

Formatting Computer Code

- All blocks of commands and computer code should be set in Courier New font at a type size that minimizes wrapping of lines (usually 9 point).
- Please use tabs and not the space bar to introduce indentations and align text.
- Filenames and paths should be set in Courier New Font at the default type size, but URLs and e-mail addresses should be set in the default font in italics.

Literature Cited

Current Protocols uses APA style. References should be listed alphabetically by author's last name. Should you wish to use EndNote or Zotero, please download the corresponding output style:

[EndNote](#)

[\(ens\)](#)

[Zotero \(csl\)](#)

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Figures, supporting information, and appendices should be supplied as separate files. You should review the [basic figure requirements](#) for manuscripts for peer review, as well as the more detailed post-acceptance figure requirements. Current Protocols does not limit the number of figures in a manuscript. There is no charge for the inclusion of color figures.

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actual figures. Authors should annotate and complete their proof review within 3 days of the receipt of this email.

Continuous Publication

Current Protocols uses Continuous Publication, a model that allows an article to be published directly into an issue. In this model there is no Early View version as the ability to publish before an issue is completed provides faster publishing of articles with final citation details. Article metadata is sent to PubMed and PMC upon article publication rather than waiting until the issue has published.

Access and Sharing

This [chart](#) summarizes Wiley's article sharing policies and how to share your research responsibly.

When the article is published online:

- The author receives an email alert (if requested).
- The link to the published article can be shared through social media.
- The author will have free access to the paper (after accepting the Terms & Conditions of use, they can view the article).
- The corresponding author and co-authors can nominate up to ten colleagues to receive a publication alert and free online access to the article.

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Appendix

Resource Identification Initiative

The journal supports the [Resource Identification Initiative](#), which aims to promote research resource identification, discovery, and reuse. This initiative, led by the [Neuroscience Information Framework](#) and the [Oregon Health & Science University Library](#), provides unique identifiers for antibodies, model organisms, cell lines, and tools including software and databases. These IDs, called Research Resource Identifiers (RRIDs), are machine-readable and can be used to search for all papers where a particular resource was used and to increase access to critical data to help researchers identify suitable reagents and tools.

You are encouraged to use RRIDs to cite the resources used in your research where applicable in the text, similar to a regular citation or Genbank Accession number. For antibodies, you should include in the citation the vendor, catalogue number, and RRID both in the text upon first mention in the Methods section. For software tools and databases, please provide the name of the resource followed by the resource website, if available, and the RRID. For model organisms, the RRID alone is sufficient.

Additionally, you should include the RRIDs in the list of keywords associated with the manuscript.

To Obtain Research Resource Identifiers (RRIDs):

- Use the [Resource Identification Portal](#), created by the Resource Identification Initiative Working Group.
- Search for the research resource (please see the section titled “Search Features and Tips” for more information).
- Click on the “Cite This” button to obtain the citation and insert the citation into the manuscript text.

If there is a resource that is not found within the [Resource Identification Portal](#), you are asked to register the resource with the appropriate resource authority. Information on how to do this is provided in the “Resource Citation Guidelines” section of the Portal.

If any difficulties in obtaining identifiers arise, please contact rii-help@scicrunch.org for assistance.

Example Citations:

Antibodies: "Wnt3 was localized using a rabbit polyclonal antibody C64F2 against Wnt3 (Cell Signaling Technology, Cat# 2721S, RRID: AB_2215411)"

Model Organisms: "Experiments were conducted in c. elegans strain SP304 (RRID:CGC_SP304)"

Cell lines: "Experiments were conducted in PC12 CLS cells (CLS Cat# 500311/p701_PC-12, RRID:CVCL_0481)"

Tools, Software, and Databases: "Image analysis was conducted with CellProfiler Image Analysis Software, V2.0 (<http://www.cellprofiler.org>, RRID:nif-0000-00280)"

Species Names

Upon its first use in the title, abstract, and text, the common name of a species should be followed by the scientific name (genus, species, and authority) in parentheses. For well-known species, however, scientific names may be omitted from article titles. If no common name exists in English, only the scientific name should be used.

Genetic Nomenclature

Sequence variants should be described in the text and tables using both DNA and protein designations whenever appropriate. Sequence variant nomenclature must follow the current HGVS guidelines; see varnomen.hgvs.org, where examples of acceptable nomenclature are provided.

Sequence Data

Nucleotide sequence data can be submitted in electronic form to any of the three major collaborative databases: DDBJ, EMBL, or GenBank. It is only necessary to submit to one database as data are exchanged between DDBJ, EMBL, and GenBank on a daily basis. The suggested wording for referring to accession-number information is: 'These sequence data have been submitted to the DDBJ/EMBL/GenBank databases under accession number U12345'. Addresses are as follows:

- DNA Data Bank of Japan (DDBJ): ddbj.nig.ac.jp
- EMBL Nucleotide Archive: ac.uk/ena
- GenBank: ncbi.nlm.nih.gov/genbank

Proteins sequence data should be submitted to either of the following repositories:

- Protein Information Resource (PIR): georgetown.edu
- SWISS-PROT: ch/sprot/sprot-top

Structural Data

For papers describing structural data, atomic coordinates and the associated experimental data should be deposited in the appropriate databank (see below). **Please note that the data in databanks must be released, at the latest, upon publication of the article.** We trust in the cooperation of our authors to ensure that atomic coordinates and experimental data are released on time.

- **Organic and organometallic compounds:** Crystallographic data should not be sent as Supporting Information but should be deposited with the *Cambridge Crystallographic Data Centre* (CCDC) at cam.ac.uk/services/structure%5Fdeposit.
- **Inorganic compounds:** *Fachinformationszentrum Karlsruhe* (FIZ; fiz-karlsruhe.de).
- **Proteins and nucleic acids:** *Protein Data Bank* (rcsb.org).
- **NMR spectroscopy data:** *BioMagResBank* (wisc.edu).

Embedded Rich Media

This journal has the option for authors to embed rich media (i.e. video and audio) within their final article. These files should be submitted with the manuscript files online, using either the “Embedded Video” or “Embedded Audio” file designation. If the video/audio includes dialogue, a transcript should be included as a separate file. **The combined manuscript files, including video, audio, tables, figures, and text must not exceed 350 MB.** For full guidance on accepted file types and resolution please see [here](#).

Ensure each file is numbered (e.g., Video 1, Video 2, etc.) Legends for the rich media files should be placed at the end of the article.

The content of the video should not display overt product advertising. Educational presentations are encouraged.

Any narration should be in English, if possible. A typed transcript of any speech within the video/audio should be provided. An English translation of any non-English speech should be provided in the transcript.

All embedded rich media will be subject to peer review. Editors reserve the right to request edits to rich media files as a condition of acceptance. Contributors are asked to be succinct, and the Editors reserve the right to require shorter video/audio duration. The video/audio should be high quality (both in content and visibility/audibility). The video/audio should make a specific point; particularly, it should demonstrate the features described in the text of the manuscript.

Participant Consent: It is the responsibility of the corresponding author to seek informed consent from any identifiable participant in the rich media files. Masking a participant’s eyes, or excluded head and shoulders is not sufficient. Please ensure that a consent form

(<https://authorservices.wiley.com/author-resources/Journal-Authors/licensing/licensing-info-faqs.html>) is provided for each participant.

APÊNDICE B — Convite para submeter o trabalho a revista



Natalia Iorio Lopes Pontes Povea <nataliaiorio@id.uff.br>

Invitation to submit to Current Protocols

1 mensagem

Current Protocols <cpsubmissions@wiley.com>
Para: Natalia Iorio <nataliaiorio@id.uff.br>

2 de setembro de 2025 às 17:00

WILEY

Dear Dr. Natalia Iorio,

I would like to invite you to submit an article to Current Protocols. Please let me know as soon as possible if you will be able to accept my invitation by clicking on either the ACCEPT or DECLINE buttons below.

I have set the submission date for six months from now. If you need more time, just let me know. Please download our template [here](#), and read our [author guidelines](#) closely and do not deviate from our required formatting as doing so will slow the review process. If you would like to see a finished product, please go to the Current Protocols website and download a protocol.

- Article Type: Protocol
- Recommended title: A Novel Laboratorial Protocol to Evaluate Bacterial Microleakage of Endodontic Sealers

Please respond to this invitation by 27 September 2025.

DECLINE

ACCEPT

Thank you in advance for considering our invitation. Please note that your manuscript would undergo the same editorial evaluation process as all other submissions to this journal and acceptance for publication is therefore not guaranteed. I look forward to hearing from you.

Kind regards,
Dr. Alexander Hutchison
Current Protocols

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APÊNDICE C— Cadastro de atividade no SISGEN



Ministério do Meio Ambiente CONSELHO DE GESTÃO DO PATRIMÔNIO GENÉTICO

SISTEMA NACIONAL DE GESTÃO DO PATRIMÔNIO GENÉTICO E DO CONHECIMENTO TRADICIONAL ASSOCIADO

Comprovante de Cadastro de Acesso

Cadastro nº A26E3E0

A atividade de acesso ao Patrimônio Genético, nos termos abaixo resumida, foi cadastrada no SisGen, em atendimento ao previsto na Lei nº 13.123/2015 e seus regulamentos.

Número do cadastro: **A26E3E0**
 Usuário: **UFF**
 CPF/CNPJ: **28.523.215/0001-06**
 Objeto do Acesso: **Patrimônio Genético**
 Finalidade do Acesso:
☒ **Pesquisa Científica** ☐ **Bioprospecção** ☐ **Desenvolvimento Tecnológico**

Espécie

Título da Atividade: **Análise fenotípica e genotípica da virulência, susceptibilidade, patogenicidade de amostras microbianas**

Equipe

Natalia Iorio Lopes Pontes Póvoa	UFF
Helvécio Cardoso Corrêa Póvoa	Universidade Federal Fluminense
Raiane Cardoso Chamon	Universidade Federal Fluminense

Data do Cadastro: **02/05/2023 22:36:29**

Situação do Cadastro: **Concluído**

Conselho de Gestão do Patrimônio Genético
 Situação cadastral conforme consulta ao SisGen em **22:36 de 02/05/2023**.



SISTEMA NACIONAL DE GESTÃO
 DO PATRIMÔNIO GENÉTICO
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